SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

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1. Introduction, Given Data and Assumptions

A head-office link carries the plant's production data over fibre. Because the link is long and fast, the choice of ARQ protocol and window size decides how much of the 1 Gbps capacity is actually used. This report calculates the timing of the link, the utilisation of Stop-and-Wait ARQ, the window and sequence-number sizes needed by Go-Back-N and Selective Repeat, explains how HDLC and PPP mark frame boundaries, repeats the window calculation for a satellite link, and evaluates a Selective Repeat window of 64.

1.1 Given data

Quantity	Head-office link	Satellite link
Distance (d)	400 km = 400,000 m	36,000 km = 36,000,000 m
Data rate (R)	1 Gbps = 10^9 bits/s	10 Mbps = 10^7 bits/s
Frame size (L)	1,500 bytes × 8 = 12,000 bits	12,000 bits
Signal speed (v)	2×10^8 m/s	2×10^8 m/s

1.2 Assumptions

- ACK transmission time is negligible.
- Processing delay at sender and receiver is negligible.
- The link is ideal and error-free for all utilisation calculations.
- **One-way** propagation delay is used for Tp; the round trip is 2·Tp.
- The standard sliding-window utilisation model is applied: U = min(1, W / (1 + 2a)).

2. Part A – Head-Office Link

2.1 Transmission time, propagation time and ratio a

```
Tt = L / R   = 12,000 bits / (1 × 10⁹ bits/s)
             = 1.2 × 10⁻⁵ s
             = 12 µs
Tp = d / v   = 400,000 m / (2 × 10⁸ m/s)
             = 2 × 10⁻³ s
             = 2 ms  (one-way)
a  = Tp / Tt = 2 ms / 0.012 ms (= 2000 µs / 12 µs)        = 166.67
1 + 2a       = 1 + 2(166.67) = 334.33
```

2.2 Stop-and-Wait ARQ utilisation and throughput

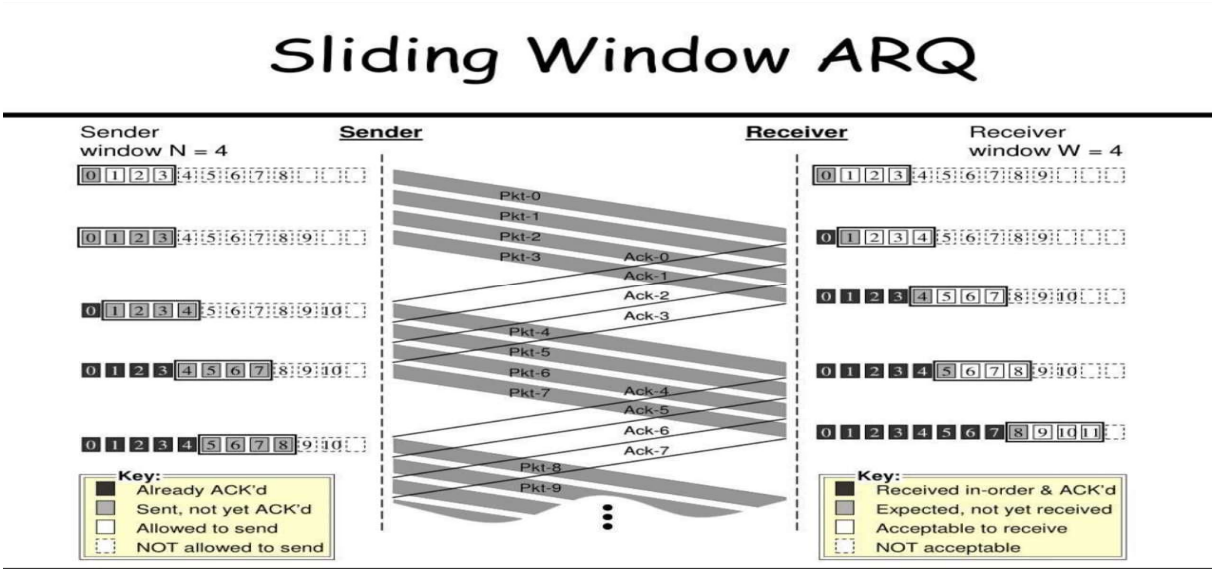
```
U = Tt / (Tt + 2Tp) = 1 / (1 + 2a)
                    = 1 / 334.33
                    = 0.002991 = 0.299 %

Cycle time = Tt + 2Tp = 12 µs + 4,000 µs = 4,012 µs
Throughput = U × R = 0.002991 × 1,000 Mbps = 2.99 Mbps ≈ 3 Mbps
Check: 12,000 bits / 4.012 ms = 2.99 Mbps
```

Why utilisation is so low. Stop-and-Wait sends one frame and then waits for its ACK. The frame takes only 12 µs to transmit, but the signal needs 2 ms to reach the receiver and another 2 ms for the ACK to return. The sender is therefore idle for about 4,000 µs of every 4,012 µs cycle. The longer and faster the link (the larger a), the more frames could have been sent during the wait, so only about 0.3 % of the 1 Gbps capacity is used.

2.3 Minimum window for 100 % utilisation

```
The sender must keep transmitting until the first ACK returns:
W ≥ 1 + 2a = 334.33
W = ceiling(334.33) = 335 frames
Check: 335 × 12 µs = 4,020 µs ≥ 4,012 µs  → link never idles
```



2.4 Sequence-number bits

Go-Back-N needs W ≤ 2^k − 1 (the receiver window is 1, so one sequence number must stay unused). Selective Repeat needs W ≤ 2^(k−1) (sender and receiver windows must not overlap). With W = 335:

k (bits)	Sequence numbers 2^k	Go-Back-N max window 2^k − 1	Selective Repeat max window 2^(k−1)	Fits W = 335?
8	256	255	128	Neither
9	512	511	256	**Go-Back-N only**
10	1,024	1,023	512	**Selective Repeat also**

Final answer: **Go-Back-N needs k = 9 bits; Selective Repeat needs k = 10 bits.**

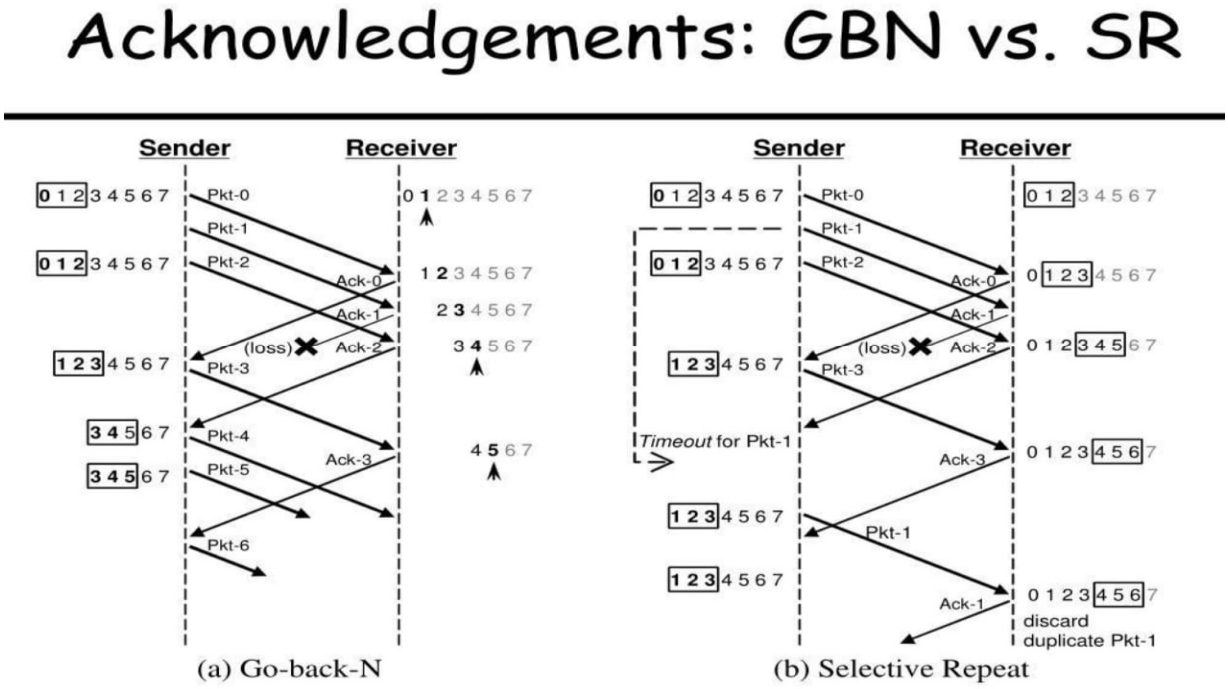
3. Part B – Go-Back-N versus Selective Repeat

Feature	Go-Back-N	Selective Repeat
Sender window	Up to 2^k − 1 (here 335 frames)	Up to 2^(k−1) (here 335 frames)
Receiver window	1 frame (accepts only the next in-order frame)	Equal to sender window (accepts frames out of order)
Frame lost	Receiver discards every later frame until the lost one arrives; sender resends the lost frame and all frames sent after it	Receiver keeps later frames in its buffer; sender resends only the lost frame
Frames retransmitted after one loss	Lost frame + all following unacknowledged frames (up to 335 here)	**1 frame**
Receiver buffer	Minimal (no out-of-order storage)	Large: up to W frames, then re-ordering before delivery
Complexity	Simple	Higher: per-frame timers/ACKs and re-ordering logic
Sequence-number bits (W = 335)	9	10

What happens after one loss. Suppose frame n is lost while frames n+1, n+2, … are already on the way. In Go-Back-N the receiver cannot accept them out of order, so it discards them; when the sender times out (or receives a NAK) it goes back and resends frame n and everything after it. In Selective Repeat the receiver stores the later frames, acknowledges them individually and waits only for frame n; the sender resends frame n alone.

Why Selective Repeat costs more. The receiver must hold up to W out-of-order frames in memory (Section 6 gives the figure), track which frames have arrived, and deliver them in order once the gap is filled. This needs more buffer memory and more complex logic than Go-Back-N's single-frame receiver.

Conclusion: Selective Repeat retransmits only the missing frame, whereas Go-Back-N retransmits the lost frame and the subsequent unacknowledged frames. On this 400 km link a single loss could waste up to 335 frames under Go-Back-N, so Selective Repeat is the better choice if the receiver memory is available.



4. Part C – HDLC and PPP Frame Boundaries

Both protocols mark the start and end of a frame with a special **flag** pattern: **01111110 (0x7E)** in HDLC and **0x7E** in PPP (the same bit pattern). The receiver looks for this pattern to find where a frame begins and ends. The general frame layout is:

Start Flag	Header	Payload / Data	FCS	End Flag
01111110 (0x7E)	Address + Control	Variable length	16- or 32-bit CRC	01111110 (0x7E)

Figure 1: General HDLC/PPP frame format (fields are sent left to right).

Field	HDLC	PPP (RFC 1662)
Start / End Flag	01111110 (0x7E)	0x7E
Header	Address (8 bits) + Control (8 bits)	Address 0xFF + Control 0x03 + Protocol (1–2 bytes)
Payload / Data	Information field (any length)	Datagram (e.g. IP packet)
FCS	16-bit CRC (32-bit optional)	16-bit CRC (32-bit optional)

4.1 Bit stuffing in HDLC

The flag 01111110 could also appear inside the data. To prevent this, the HDLC sender inserts a 0 after every five consecutive 1s in the data (flags excluded). The receiver removes a 0 that follows five consecutive 1s, so the flag never appears inside a frame body.

```
Data to send     : 0 1 1 1 1 1 1 0     (looks like a flag)
After stuffing   : 0 1 1 1 1 1 0 1 0   (0 inserted after the five 1s)
Receiver removes the 0 following five 1s → original data restored
```

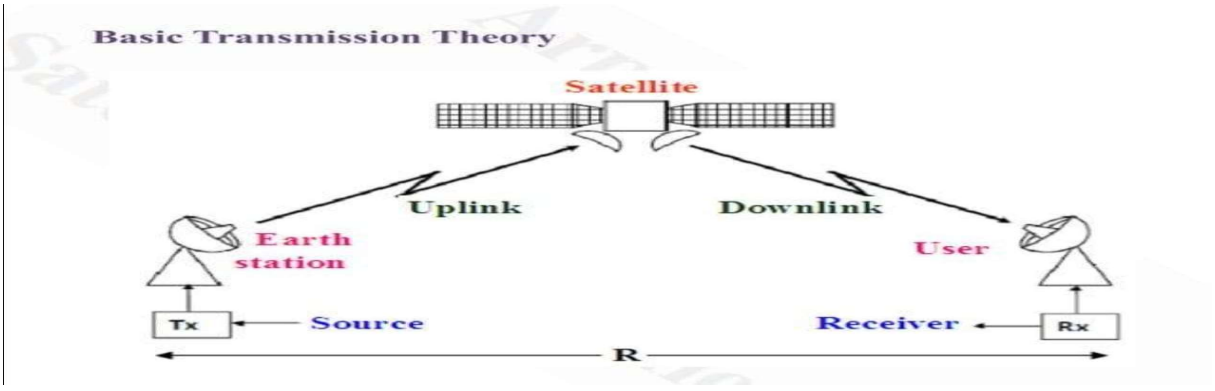
4.2 Byte stuffing and bit-oriented framing in PPP

Asynchronous PPP (byte-oriented links such as serial lines) uses **byte stuffing**. The escape byte is 0x7D. If 0x7E or 0x7D appears in the data, the sender replaces it by 0x7D followed by the original byte XOR 0x20: 0x7E becomes **0x7D 0x5E** and 0x7D becomes **0x7D 0x5D**. The receiver reverses this. **Synchronous PPP** (bit-oriented links) uses the same **bit stuffing** as HDLC: a 0 is inserted after five consecutive 1s.

5. Part D – Satellite Link

```
Tt = L / R   = 12,000 bits / (1 × 10⁷ bits/s)   = 1.2 × 10⁻³ s = 1.2 ms
Tp = d / v   = 36,000,000 m / (2 × 10⁸ m/s)     = 0.18 s        = 180 ms  (one-way)
a  = Tp / Tt = 180 ms / 1.2 ms                  = 150
W ≥ 1 + 2a   = 1 + 2(150) = 301 frames
Check: Stop-and-Wait U = 1/301 = 0.33 %
```

Why a large window is needed. The round trip takes 2 × 180 = 360 ms, while one frame takes only 1.2 ms to transmit. During one round trip the sender can send 300 further frames after the first one, so it must be allowed to have 301 frames outstanding before the first ACK returns. A small window would leave the satellite channel idle for most of every round trip.



6. Part E – Alternative Concept Analysis: Selective Repeat with W = 64

For the head-office link, suppose Selective Repeat uses a window of **64** frames instead of the required 335 (1 + 2a = 334.33).

```
Utilisation = W / (1 + 2a) = 64 / 334.33
            = 0.1914
            = 19.14 %
  (check: 64 × 12 µs = 768 µs sent per 4,012 µs cycle = 19.14 %)
Throughput  = U × R = 0.1914 × 1,000 Mbps
                    = 191.4 Mbps
Receiver buffer = 64 frames × 1,500 bytes
                = 96,000 bytes = 96 kB (decimal)
```

The receiver must be able to buffer up to 64 frames, excluding extra implementation overhead (headers, per-frame state, re-ordering structures). With W = 64 the link carries 191.4 Mbps, about 64 times more than Stop-and-Wait (2.99 Mbps) but still only 19.14 % of capacity.

Design	Window	Utilisation	Throughput	Receiver buffer
Stop-and-Wait	1	0.299 %	2.99 Mbps	1,500 bytes
Selective Repeat, W = 64	64	19.14 %	191.4 Mbps	96,000 bytes (96 kB)
Selective Repeat, W = 335	335	100 %	1,000 Mbps	502,500 bytes (502.5 kB)

Justified protocol choice. For the head-office link, Selective Repeat with W = 335 (k = 10 bits) fully uses the 1 Gbps link and, after a loss, resends only one frame; its cost is about 502.5 kB of receiver buffer, which is small for modern equipment. A window of 64 would save buffer memory but waste about 81 % of the capacity.

7. Summary of Results

Item	Result
Transmission time (head office)	12 µs
One-way propagation time	2 ms
Ratio a = Tp / Tt	166.67
Stop-and-Wait utilisation / throughput	0.299 % / 2.99 Mbps (≈ 3 Mbps)
Minimum window (100 % utilisation)	335 frames
Sequence-number bits	Go-Back-N: 9 bits; Selective Repeat: 10 bits
Fewer retransmissions after one loss	Selective Repeat (1 frame vs. lost frame + following frames)
HDLC / PPP flag	01111110 = 0x7E; PPP flag 0x7E
Satellite link: Tt, Tp, a, minimum window	1.2 ms, 180 ms, 150, 301 frames
Selective Repeat, W = 64: utilisation / throughput / buffer	19.14 % / 191.4 Mbps / 96,000 bytes (96 kB)

8. Conclusion

On the 400 km, 1 Gbps head-office link, one-way propagation (2 ms) is 166.67 times the frame transmission time (12 µs), so Stop-and-Wait uses only 0.299 % of the link (2.99 Mbps). A sliding window of at least 335 frames is needed for full utilisation, requiring 9-bit sequence numbers for Go-Back-N or 10-bit for Selective Repeat. Selective Repeat retransmits only the lost frame at the cost of more receiver buffer, and a window of 64 would give just 191.4 Mbps. HDLC and PPP delimit frames with the 0x7E flag, using bit stuffing (HDLC, synchronous PPP) or byte stuffing (asynchronous PPP). The satellite link needs a window of 301 frames because of its 180 ms one-way delay.